Apache Flink Kickstarter II **[UNDER CONSTRUCTION]**

> ***On GitHub, Watch → Custom → Releases is the most useful setting if you want to stay updated on when something important is released for this project.***

Apache Flink Kickstarter II is the 2026 evolution of my original Kickstarter project — rebuilt to showcase the cutting edge of **Apache Flink 2.1.x**.

Designed as a hands-on, production-minded accelerator, it brings Flink to life *locally* on **Confluent Platform on Minikube**, while drawing direct comparisons to **Confluent Cloud for Apache Flink** — so you can clearly see what's possible across environments.

Every **example** is delivered end-to-end — from schema design to fully operational streaming pipelines — with implementations in **both Java and Python (when possible)** where it matters, bridging real-world developer workflows with modern streaming architecture.

Table of Contents

- 1.0 Prerequisites
  - 1.1 Confluent Platform on Minikube — Production-Like Streaming, Running Locally
    - 1.1.1 Requirements
    - 1.1.2 Local Infrastructure Deployment Made Simple with a `Makefile`
      - 1.1.2.1 Install the Tooling (One Command Setup)
      - 1.1.2.2 Full stack (CP + Kafka UI)
      - 1.1.2.3 Add Apache Flink + CMF (run separately after `make cp-up`)
    - 1.1.3 `Makefile` Composite Workflow Target Reference
    - 1.1.4 `Makefile` Individual Target Reference
    - 1.1.5 `Makefile` Target Configuration Reference
    - 1.1.6 Remote Server Setup (SSH Tunneling)
  - 1.2 Confluent Cloud Setup
    - 1.2.1 Requirements
- 2.0 The Examples
  - 2.1 Apache Flink User-Defined Functions (UDF)
    - 2.1.1 Process Table Functions (PTF)
      - 2.1.1.1 Limitation(s)
- 3.0 Debugging the Examples
  - 3.1 Apache Flink UDF Debugging with Java Debug Wire Protocol (JDWP)
    - 3.1.1 Process Table Functions (PTF)
      - 3.1.1.1 Debugging the `row-driven` PTF using set semantics (`UserEventEnricher`)
      - 3.1.1.2 Debugging the `row-driven` PTF using row semantics (`OrderLineExpander`)

- 3.1.1.3 Debugging the `timer-driven` PTFs (`SessionTimeoutDetector`, `AbandonedCartDetector`, `PerEventFollowUp`, and `SlaMonitor`)
- 4.0 Resources
  - 4.1 Confluent for Kubernetes (CfK)
  - 4.2 Confluent Platform for Apache Flink
  - 4.3 Confluent Cloud for Apache Flink (CCAF)

1.0 Prerequisites

1.1 Confluent Platform on Minikube — Production-Like Streaming, Running Locally

To ***run***, ***test***, and ***debug*** Apache Flink like a production engineer, this project provides a full Confluent Platform stack running locally on Minikube — no cloud required.

You get a production-like environment on your laptop:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.1** via the Confluent Flink Kubernetes Operator 1.130
- **Confluent Manager for Apache Flink (CMF) 2.1** for Flink environment management
- **Kafka UI** (Provectus) for cluster inspection

The included `Makefile` acts as your control plane—automating setup, teardown, and day-to-day workflows—so you can focus on building Flink pipelines, not infrastructure.

> Build locally. Debug with confidence. Deploy to production-ready environments.

1.1.1 Requirements

To run this project, you'll need **macOS (with Homebrew)** or **Linux (with apt-get)**.

The full stack — **Minikube + Confluent Platform + Flink + CMF + Kafka UI** — is resource-intensive and designed to mirror a production-like environment. The following defaults are recommended:

Resource	Default
CPUs	6
Memory	20 GB
Disk	50 GB

> These settings ensure stable performance across all components. You can tune them if needed, but lower resources may lead to pod restarts or degraded performance.

1.1.2 Local Infrastructure Deployment Made Simple with a `Makefile`

`Makefile` Architecture

```
graph TD
    %% — Composite entry points
```

```
————————————————————————————————————————————
    CP_UP(["`**make cp-up**`"])
    FLINK_UP(["`**make flink-up**`"])
    CP_DOWN(["`**make cp-down**`"])
    FLINK_DOWN(["`**make flink-down**`"])
    TEARDOWN(["`**make confluent-teardown**`"])
    NUKE(["`**make nuke**`"])

    %% ── Phase 1: Prerequisites
————————————————————————————————————————————
    subgraph P1["Phase 1 — Prerequisites"]
        INSTALL_PRE["install-prereqs\ndocker · kubectl · minikube\nhelm ·
gettext · gradle · openjdk-21"]
        CHECK_PRE["check-prereqs\ndocker · kubectl · minikube · helm ·
java 21"]
        UNINSTALL_PRE["uninstall-prereqs\nremove all installed tooling"]
    end

    %% ── Phase 2: Minikube
————————————————————————————————————————————————
    subgraph P2["Phase 2 — Minikube"]
        MK_START["minikube-start\ncpus=6 · mem=20GB · disk=50GB"]
        MK_STOP["minikube-stop"]
        MK_DELETE["minikube-delete"]
    end

    %% ── Phase 3: Confluent Operator
————————————————————————————————————————
    subgraph P3["Phase 3 — Confluent Operator"]
        NS["namespace\nkubectl create namespace confluent"]
        OP_INSTALL["operator-install\nhelm: confluent-for-kubernetes"]
        OP_UNINSTALL["operator-uninstall"]
    end

    %% ── Phase 4: Confluent Platform
————————————————————————————————————————
    subgraph P4["Phase 4 — Confluent Platform"]
        CP_DEPLOY["cp-deploy\nKafka KRaft · SR · Connect\nksqlDB · REST
Proxy · C3"]
        CP_DELETE["cp-delete"]
    end

    %% ── Phase 5: Control Center
————————————————————————————————————————————
    subgraph P5["Phase 5 — Control Center"]
        C3["c3-open\nlocalhost:9021"]
    end

    %% ── Phase 6: Apache Flink
————————————————————————————————————————————
    subgraph P6["Phase 6 — Apache Flink"]
        CERT["flink-cert-manager\ncert-manager v1.18.2"]
        FL_OP["flink-operator-install\nhelm: confluentinc/flink-
kubernetes-operator 1.130.0"]
```

```
        FL_RBAC["flink-rbac\nsupplemental Role + RoleBinding\n(services
access for flink SA)"]
        FL_DEPLOY["flink-deploy\nenvsubst → FlinkDeployment CR\ncp-
flink:2.1.1-cp1"]
        FL_UI["flink-ui\nlocalhost:8081"]
        FL_DELETE["flink-delete"]
        FL_OP_UN["flink-operator-uninstall"]
        CERT_UN["cert-manager-uninstall"]
    end

    %% ── Phase 7: CMF
──────────────────────────────────────────────────────
    subgraph P7["Phase 7 — Confluent Manager for Apache Flink (CMF)"]
        CMF_INSTALL["cmf-install\nhelm: confluent-manager-for-apache-flink
2.1.0"]
        CMF_ENV["cmf-env-create\nPOST /cmf/api/v1/environments"]
        CMF_OPEN["cmf-open\nlocalhost:8080/cmf/api/v1/environments"]
        CMF_PROXY["cmf-proxy-inject\nsocat sidecar → C3 Flink tab"]
        CMF_UN["cmf-uninstall"]
    end

    %% ── Phase 8: Kafka UI
──────────────────────────────────────────────────
    subgraph P8["Phase 8 — Kafka UI"]
        KUI_INSTALL["kafka-ui-install\nhelm: provectus/kafka-
ui\nbootstrap: kafka:9071"]
        KUI_OPEN["kafka-ui-open\nlocalhost:8080"]
        KUI_UN["kafka-ui-uninstall"]
    end

    %% ── Phase 9: Build & Deploy PTF UDFs
──────────────────────────────────
    subgraph P9["Phase 9 — Build & Deploy PTF UDFs"]
        BUILD_PTF_SD["build-ptf-udf-row-driven\n./gradlew clean
shadowJar\n(2 UDFs in one JAR)"]
        DEPLOY_CP_PTF_SD["deploy-cp-ptf-udf-row-driven\ncopy JAR → Flink
SQL Client\n(2 row-driven pipelines)"]
        TEARDOWN_CP_PTF_SD["teardown-cp-ptf-udf-row-driven"]
        DEPLOY_CC_PTF_SD["deploy-cc-ptf-udf-row-driven\nJAR → Confluent
Cloud\n(2 row-driven pipelines)"]
        TEARDOWN_CC_PTF_SD["teardown-cc-ptf-udf-row-driven"]
        PRODUCE_UE["produce-user-events-record\nkafka-console-producer →
user_events"]
        PRODUCE_OR["produce-orders-record\nkafka-console-producer →
orders"]
        BUILD_PTF_TD["build-ptf-udf-timer-driven\n./gradlew clean
shadowJar\n(4 UDFs in one JAR)"]
        DEPLOY_CP_PTF_TD["deploy-cp-ptf-udf-timer-driven\ncopy JAR → Flink
SQL Client\n(4 timer-driven pipelines)"]
        TEARDOWN_CP_PTF_TD["teardown-cp-ptf-udf-timer-driven"]
        DEPLOY_CC_PTF_TD["deploy-cc-ptf-udf-timer-driven\nJAR → Confluent
Cloud\n(4 timer-driven pipelines)"]
        TEARDOWN_CC_PTF_TD["teardown-cc-ptf-udf-timer-driven"]
        PRODUCE_UA["produce-user-activity-record\nkafka-console-producer →
```

```
user_activity"]
        PRODUCE_UAC["produce-user-actions-record\nkafka-console-producer →
user_actions"]
        PRODUCE_SR["produce-service-requests-record\nkafka-console-
producer → service_requests"]
        PRODUCE_CE["produce-cart-events-record\nkafka-console-producer →
cart_events"]
    end

    %% ── Manifests / Templates
──────────────────────────────────────────────
    subgraph FS["k8s/base/"]
        MANIFEST[("flink-basic-deployment.yaml\nFLINK_IMAGE ·
FLINK_VERSION")]
        RBAC_MANIFEST[("flink-rbac.yaml\nRole + RoleBinding")]
    end

    %% ── make cp-up dependency chain
────────────────────────────────────────
    CP_UP --> CHECK_PRE
    CP_UP --> MK_START
    CP_UP --> CP_CORE_UP
    CP_UP --> KUI_INSTALL

    CP_CORE_UP["cp-core-up"] --> OP_INSTALL
    CP_CORE_UP --> CP_DEPLOY
    OP_INSTALL --> NS

    %% ── make flink-up dependency chain
──────────────────────────────────────
    FLINK_UP --> CERT
    FLINK_UP --> FL_OP
    FLINK_UP --> CMF_INSTALL
    FLINK_UP --> CMF_ENV
    FLINK_UP --> FL_DEPLOY
    FL_OP --> NS
    FL_DEPLOY --> FL_RBAC
    FL_DEPLOY --> MANIFEST
    FL_RBAC --> RBAC_MANIFEST

    %% ── make deploy-cp-ptf-udf-row-driven dependency chain
──────────────────────────────
    DEPLOY_CP_PTF_SD --> BUILD_PTF_SD

    %% ── make deploy-cc-ptf-udf-row-driven dependency chain
──────────────────────────────
    DEPLOY_CC_PTF_SD --> BUILD_PTF_SD

    %% ── make deploy-cp-ptf-udf-timer-driven dependency chain
──────────────────────────────
    DEPLOY_CP_PTF_TD --> BUILD_PTF_TD

    %% ── make deploy-cc-ptf-udf-timer-driven dependency chain
──────────────────────────────
```

```
    DEPLOY_CC_PTF_TD --> BUILD_PTF_TD

    %% ── make cp-down dependency chain
────────────────────────────────────────
    CP_DOWN --> KUI_UN
    CP_DOWN --> CP_DELETE
    CP_DOWN --> OP_UNINSTALL

    %% ── make flink-down dependency chain
──────────────────────────────────────
    FLINK_DOWN --> FL_DELETE
    FLINK_DOWN --> CMF_UN
    FLINK_DOWN --> FL_OP_UN
    FLINK_DOWN --> CERT_UN

    %% ── make confluent-teardown
────────────────────────────────────────────
    TEARDOWN -->|"1 — check minikube running"| FLINK_DOWN
    TEARDOWN -->|"2"| CP_DOWN
    TEARDOWN -->|"3 — delete namespace"| NS
    TEARDOWN -->|"4"| MK_STOP

    %% ── make nuke
──────────────────────────────────────────────────────────
    NUKE -->|"1"| TEARDOWN
    NUKE -->|"2"| MK_DELETE
    NUKE -->|"3"| UNINSTALL_PRE

    %% ── UI access
────────────────────────────────────────────────────────────
    CP_DEPLOY -.->|"once Running"| C3
    FL_DEPLOY -.->|"once Running"| FL_UI
    CMF_INSTALL -.->|"once Running"| CMF_OPEN
    CMF_INSTALL -.->|"C3 Flink tab"| CMF_PROXY
    KUI_INSTALL -.->|"once Running"| KUI_OPEN

    %% ── Styles
──────────────────────────────────────────────────────────────
    classDef entry    fill:#1a1a2e,stroke:#e94560,color:#fff,font-
weight:bold
    classDef install  fill:#16213e,stroke:#0f3460,color:#a8dadc
    classDef remove   fill:#2d1b1b,stroke:#8b0000,color:#ffb3b3
    classDef ui       fill:#1b2d1b,stroke:#2d6a2d,color:#b3ffb3
    classDef file     fill:#2d2b1b,stroke:#8b7500,color:#ffe680
    classDef composite fill:#2a1a2e,stroke:#9b59b6,color:#dbb8ff

    class CP_UP,FLINK_UP,CP_DOWN,FLINK_DOWN,TEARDOWN,NUKE entry
    class
CP_CORE_UP,DEPLOY_CP_PTF_SD,DEPLOY_CC_PTF_SD,DEPLOY_CP_PTF_TD,DEPLOY_CC_PT
F_TD composite
    class
INSTALL_PRE,CHECK_PRE,MK_START,NS,OP_INSTALL,CP_DEPLOY,CERT,FL_OP,FL_RBAC,
FL_DEPLOY,CMF_INSTALL,CMF_ENV,CMF_PROXY,KUI_INSTALL,BUILD_PTF_SD,BUILD_PTF
_TD,PRODUCE_UE,PRODUCE_OR,PRODUCE_UA,PRODUCE_UAC,PRODUCE_SR,PRODUCE_CE
```

```
install
    class
MK_STOP,MK_DELETE,OP_UNINSTALL,CP_DELETE,FL_DELETE,FL_OP_UN,CERT_UN,CMF_UN
,KUI_UN,UNINSTALL_PRE,TEARDOWN_CP_PTF_SD,TEARDOWN_CC_PTF_SD,TEARDOWN_CP_PT
F_TD,TEARDOWN_CC_PTF_TD remove
    class C3,FL_UI,CMF_OPEN,KUI_OPEN ui
    class MANIFEST,RBAC_MANIFEST file
```

1.1.2.1 Install the Tooling (One Command Setup)

Get everything you need to deploy the full stack locally with a single command:

```
make install-prereqs
```

This installs `docker`, `kubernetes-cli`, `minikube`, `helm`, `gettext`, `gradle`, and `openjdk-21`, via Homebrew (macOS) or apt-get (Linux). Once complete, **launch Docker Desktop** before proceeding.

1.1.2.2 Full stack (CP + Kafka UI)

```
make cp-up
```

This runs: `check-prereqs` → `minikube-start` → `namespace` → `operator-install` → `cp-deploy` → `kafka-ui-install`.

Once pods are up, open Control Center:

```
make c3-open        # http://localhost:9021
```

1.1.2.3 Add Apache Flink + CMF (run separately after `make cp-up`)

```
make flink-up
```

This runs: `flink-cert-manager` → `flink-operator-install` → `cmf-install` → `cmf-env-create` → `flink-deploy`. `flink-up` is self-contained and can also be run standalone on a fresh cluster.

Once the Flink JobManager pod is running:

```
make flink-ui       # http://localhost:8081 (background — returns prompt)
make flink-ui-stop  # stop the background port-forward
```

```
make cmf-open       # http://localhost:8080/cmf/api/v1/environments
```

To expose the Flink tab inside Control Center, inject the CMF proxy sidecar:

```
make cmf-proxy-inject
```

1.1.3 `Makefile` Composite Workflow Target Reference

Target	What it does
`make cp-up`	Full stack: Minikube + CP + Kafka UI
`make flink-up`	cert-manager + Confluent Flink Operator + CMF + Flink cluster
`make cp-down`	Remove CP, Kafka UI, and Operator (Minikube keeps running)
`make flink-down`	Remove Flink cluster, CMF, Operator, and cert-manager
`make confluent-teardown`	Full teardown: everything + stop Minikube
`make nuke`	Full wipe: confluent-teardown + minikube-delete + uninstall-prereqs (leaves machine as close to factory as possible)

1.1.4 `Makefile` Individual Target Reference

▶ Phase 1 — Prerequisites

Target	Description
`install-prereqs`	Install Docker Desktop, kubectl, Minikube, Helm, envsubst, and Gradle via Homebrew
`check-prereqs`	Verify all required tools are available
`uninstall-prereqs`	Uninstall all prerequisites installed by `install-prereqs` (safe to run even if not installed)

▶ Phase 2 — Minikube

Target	Description
`minikube-start`	Start Minikube with configured resources
`minikube-status`	Show Minikube and node status
`minikube-stop`	Stop the Minikube cluster
`minikube-delete`	Permanently delete the Minikube cluster

▶ Phase 3 — Confluent Operator

Target	Description

Target	Description
`namespace`	Create the `confluent` namespace and set it as default context
`operator-install`	Add Confluent Helm repo and install CFK Operator
`operator-status`	Show CFK Operator pod status
`operator-uninstall`	Remove the CFK Operator Helm release

▶ Phase 4 — Confluent Platform

Target	Description
`cp-deploy`	Deploy Kafka (KRaft), Schema Registry, Connect, ksqlDB, REST Proxy, Control Center
`cp-watch`	Watch pod startup live (Ctrl+C to exit)
`cp-status`	Show current pod status
`cp-delete`	Remove all CP components and leftover PVCs

▶ Phase 5 — Control Center

Target	Description
`c3-open`	Port-forward Control Center in the background and open `http://localhost:9021` (`make c3-stop` to kill)
`c3-stop`	Stop the background Control Center port-forward

▶ Phase 6 — Apache Flink

Target	Description
`flink-cert-manager`	Install cert-manager (Confluent Flink Operator dependency)
`flink-operator-install`	Install the Confluent Flink Kubernetes Operator (`confluentinc/flink-kubernetes-operator`)
`flink-operator-status`	Show Flink Operator pod status
`flink-operator-uninstall`	Remove the Confluent Flink Operator Helm release
`flink-rbac`	Apply supplemental RBAC so the `flink` SA can read services (needed for job submission)
`flink-deploy`	Deploy the Flink session cluster using `FLINK_MANIFEST` (runs `flink-rbac` first)
`flink-status`	Show Flink pods and FlinkDeployment CRs
`flink-ui`	Port-forward Flink UI in the background and open `http://localhost:8081` (`make flink-ui-stop` to kill)
`flink-ui-stop`	Stop the background Flink UI port-forward

Target	Description
`flink-delete`	Delete the Flink session cluster
`cert-manager-uninstall`	Remove cert-manager

▶ Phase 7 — Confluent Manager for Apache Flink (CMF)

Target	Description
`cmf-install`	Install CMF via Helm (`confluent-manager-for-apache-flink`) and wait for pod readiness
`cmf-env-create`	Create a Flink environment (`CMF_ENV_NAME`) in CMF pointing to the `confluent` namespace
`cmf-status`	Show CMF pod status and list registered Flink environments
`cmf-open`	Port-forward CMF REST API and open `http://localhost:8080/cmf/api/v1/environments`
`cmf-uninstall`	Uninstall CMF (safe to run even if not installed)
`cmf-proxy-inject`	Patch the C3 StatefulSet with a `socat` sidecar to expose the Flink tab in Control Center
`cmf-proxy-remove`	Remove the CMF proxy sidecar and resume CFK reconciliation
`cmf-proxy-logs`	Stream logs from the `cmf-proxy` sidecar in the C3 pod

▶ Phase 8 — Kafka UI (Provectus)

Target	Description
`kafka-ui-install`	Install Kafka UI connected to the local CP cluster (Kafka + Schema Registry + Connect)
`kafka-ui-status`	Show Kafka UI pod status
`kafka-ui-open`	Port-forward Kafka UI and open `http://localhost:8080`
`kafka-ui-uninstall`	Remove Kafka UI

▶ Phase 9 — Build & Deploy Flink JARs

Target	Description
`build-ptf-udf-row-driven`	Build the `row-driven` PTF UDF uber JAR (requires Gradle)
`deploy-cp-ptf-udf-row-driven`	Build `row-driven` UDF JAR, copy to Flink pods, and submit SQL via Flink SQL Client
`teardown-cp-ptf-udf-row-driven`	Tear down the `row-driven` PTF UDF deployment

Target	Description
`deploy-cc-ptf-udf-row-driven`	Build and deploy the `row-driven` UDF JAR to Confluent Cloud via Terraform
`teardown-cc-ptf-udf-row-driven`	Tear down the `row-driven` PTF UDF deployment from Confluent Cloud
`build-ptf-udf-timer-driven`	Build the `timer-driven` PTF UDF uber JAR (requires Gradle)
`deploy-cp-ptf-udf-timer-driven`	Build `timer-driven` UDF JAR, copy to Flink pods, and submit SQL via Flink SQL Client
`teardown-cp-ptf-udf-timer-driven`	Tear down the `timer-driven` PTF UDF deployment
`deploy-cc-ptf-udf-timer-driven`	Build and deploy the `timer-driven` UDF JAR to Confluent Cloud via Terraform
`teardown-cc-ptf-udf-timer-driven`	Tear down the `timer-driven` PTF UDF deployment from Confluent Cloud

1.1.5 `Makefile` Target Configuration Reference

All variables are overridable at the command line.

▶ Defaults

Variable	Default	Description
`NAMESPACE`	`confluent`	Kubernetes namespace
`CONFLUENT_MANIFEST`	`k8s/base/confluent-platform-c3++.yaml`	Path to Confluent Platform manifest
`MINIKUBE_CPUS`	`6`	vCPUs allocated to Minikube
`MINIKUBE_MEM`	`20480`	Memory in MB
`MINIKUBE_DISK`	`50g`	Disk size
`FLINK_OPERATOR_VER`	`1.130.0`	Confluent Flink Kubernetes Operator version
`FLINK_IMAGE`	`confluentinc/cp-flink:2.1.1-cp1-java21` (auto-selects `-arm64` suffix on arm64 nodes)	Flink container image
`FLINK_VERSION`	`v2_1`	Flink API version string for the FlinkDeployment CR
`FLINK_CLUSTER_NAME`	`flink-basic`	Name of the FlinkDeployment resource

Variable	Default	Description
`FLINK_MANIFEST`	`k8s/base/flink-basic-deployment.yaml`	Path to FlinkDeployment template
`FLINK_RBAC_MANIFEST`	`k8s/base/flink-rbac.yaml`	Path to supplemental RBAC manifest for the `flink` ServiceAccount
`CERT_MANAGER_VER`	`v1.18.2`	cert-manager version
`CMF_VER`	`2.1.0`	Confluent Manager for Apache Flink version
`CMF_PORT`	`8080`	CMF REST API local port
`CMF_ENV_NAME`	`dev-local`	Flink environment name registered in CMF
`C3_PORT`	`9021`	Control Center local port
`FLINK_UI_PORT`	`8081`	Flink UI local port
`KAFKA_UI_PORT`	`8080`	Kafka UI local port
`PTF_UDF_TOPICS`	`user_events enriched_events`	Kafka topics for the ptf_udf Flink job

> **Note:** CMF uses the Confluent-packaged Flink operator (`confluentinc/flink-kubernetes-operator`) and `confluentinc/cp-flink` images — not the Apache OSS Flink operator or `flink` Docker Hub image.

Example — deploy a specific Flink image:

```
make flink-deploy FLINK_IMAGE=confluentinc/cp-flink:2.1.1-cp1-java21-arm64
FLINK_VERSION=v2_1
```

1.1.6 Remote Server Setup (SSH Tunneling)

If the full stack is running on a remote server (e.g., a Vultr VPS), you need two things: a terminal on the remote to run `make` targets, and an SSH tunnel to reach the UIs from your local browser.

▶ Step 0 — Authorize your local machine's SSH public key on the remote server

On the remote server, append your local machine's public key to the authorized keys file so you can connect without a password:

```
echo "your-public-key-string" >> ~/.ssh/[file-name]
```

Replace `your-public-key-string` with the contents of your local `~/.ssh/[file-name].pub` (e.g., `cat ~/.ssh/ssh-key-dev-cloud-server-access.pub` on your Mac), and `[file-name]` with the name of your authorized keys file (typically `authorized_keys`).

For example:

```
echo "ssh-ed25519 AAAA...your-key... user@macbook" >>
~/.ssh/authorized_keys
```

Then make sure permissions are correct on the remote:

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

▶ Step 1 — Add an entry to `~/.ssh/config` on your local machine

```
Host [label for the remote server]
  HostName [IP address]
  User root
  IdentityFile ~/.ssh/ssh-key-dev-cloud-server-access
  IdentitiesOnly yes
  LocalForward 9021 localhost:9021
  LocalForward 8081 localhost:8081
  LocalForward 8080 localhost:8080
```

The three `LocalForward` lines tunnel the UI ports from the remote server to your local machine automatically every time you connect.

▶ Step 2 — Terminal 1: bring up the stack on the remote

```
ssh [label for the remote server]
cd /path/to/apache_flink-kickstarter-ii
make cp-up
make flink-up
```

▶ Step 3 — Terminal 2: open the SSH tunnel

```
ssh [label for the remote server]
```

Connecting activates the `LocalForward` rules. No extra flags needed.

▶ Step 4 — Open the UIs in your local browser

URL	UI
`http://localhost:9021`	Confluent Control Center
`http://localhost:8081`	Apache Flink UI
`http://localhost:8080`	Kafka UI

> The port-forwards on the remote side are started by `make c3-open`, `make flink-ui`, and `make kafka-ui-open` (called internally by `make cp-up` / `make flink-up`). The SSH `LocalForward` simply bridges your laptop to those already-listening ports on the remote.

1.2 Confluent Cloud Setup

1.2.1 Requirements

Before you begin, ensure you have access to the following cloud accounts:

- **Confluent Cloud Account** — for Kafka, Schema Registry, and Flink resources
- **Terraform Cloud Account** — for automated infrastructure provisioning

Make sure the following tools are installed on your local machine:

- **Java JDK 21** — for building Flink UDFs
- **Gradle 9.4.1 or higher** — for building Flink UDFs
- **Terraform CLI version 1.13.0 or higher** — for deploying infrastructure to Confluent Cloud

---

2.0 The Examples

Once you've set up **Confluent Platform on Minikube** or created your **Confluent Cloud** account, you're ready to try the examples.

2.1 Apache Flink User-Defined Functions (UDF)

2.1.1 Process Table Functions (PTF)

▶ ***What are PTFs?***

PTFs are a special type of Apache Flink UDF that offers stateful, timer-aware processing capabilities directly within Flink SQL. PTFs can be either `row-driven` (invoked for each input row) or `timer-driven` (triggered based on timers you set in your code).

▶ ***Why use PTFs?***

PTF UDFs are ideal when you need memory across rows, respond to time passing — not just on arriving data, or you want to implement complex event processing (CEP) patterns that are difficult to express in pure Flink SQL.

▶ ***When do you use PTFs?***

PTFs are used when your use case requires state and/or timers that go beyond what standard Flink SQL can handle. For example: **Stateful Enrichment with or without External Lookups**, **Per-Row Stateful Transformation**, **Complex Conditional Routing and/or Filtering**, **Session timeout**, **Abandoned cart**, **Device heartbeat monitoring**, **Per-event follow-up**, **SLA monitoring**, **Delayed side-effects**, and more.

▶ ***Where do you use PTFs?***

You write PTF UDFs as Java classes, deploy them as JAR files, and run them within your Flink SQL queries.

▼ ***How are examples of PTFs put into practice?***

Type	Purpose	Confluent Platform on Minikube	Confluent Cloud
PTF UDF-type (`row-driven`)	Walks through both **local** and cloud environments *building*, *deploying*, and *testing* two **`row-driven` PTF UDFs** (no timers) bundled in one JAR that illustrate both `ArgumentTrait` modes: **User Event Enricher** (`SET_SEMANTIC_TABLE` — enriches Kafka user events with per-user session tracking using keyed state) and **Order Line Expander** (`ROW_SEMANTIC_TABLE` — stateless one-to-many expansion of an order row into individual line items).	`CP Deploy`	`CC Deploy`
PTF UDF-type (`timer-driven`)	Walks through both **local** and cloud environments *building*, *deploying*, and *testing* four **`timer-driven` PTF UDFs** bundled in one JAR: **Session Timeout Detector** (named timers using the inactivity pattern), **Abandoned Cart Detector** (named timers using the inactivity pattern for e-commerce), **Per-Event Follow-Up** (unnamed timers using the scheduling pattern), and **SLA Monitor** (unnamed timers using the scheduling pattern).	`CP Deploy`	`CC Deploy`

> ⏳ **PTF Rules of Thumb: The Hourglass Pattern**

2.1.1.1 Limitation(s)

- Why `@StateHint` POJO with `Map` or `List` Are Sensitive to "Extremely Large State"

3.0 Debugging the Examples

3.1 Apache Flink UDF Debugging with Java Debug Wire Protocol (JDWP)

You can attach your IDE's debugger (VS Code or IntelliJ IDEA) to a running Flink TaskManager and *hit breakpoints inside your UDF code* — even though it's executing on a remote Java Virtual Machine (JVM) inside Kubernetes. The `FlinkDeployment` Custom Resource (CR) already has **Java Debug Wire Protocol (JDWP)** enabled, and debug configurations are pre-wired for both VS Code and IntelliJ IDEA.

Prerequisites: The Confluent Platform and Flink stack must be running (`make cp-up && make flink-up`), and your UDF must be deployed.

3.1.1 Process Table Functions (PTF)

3.1.1.1 Debugging the `row-driven` PTF using set semantics (`UserEventEnricher`)

> For the full deep-dive, see Remote Debugging Row-Driven Flink PTF UDFs. The same guide also covers debugging `OrderLineExpander` (§1.3).

Deploy first: `make deploy-cp-ptf-udf-row-driven`, and then:

▶ 1. Set a breakpoint

Open `UserEventEnricher.java` and click in the gutter at the first line of the `eval()` method:

```
String eventType = input.getFieldAs("event_type");
```

▶ 2. Attach the debugger

Select the **"Attach to Flink TaskManager (Row-Driven)"** configuration and start debugging. The IDE will automatically port-forward to the TaskManager pod and attach to the JDWP agent on port `5005`.

- **VS Code:** Open the **Run and Debug** panel (⇧⌘D), select the configuration from the dropdown, and press **F5**
- **IntelliJ IDEA:** Open the **Run/Debug Configurations** dropdown (top-right toolbar), select the configuration, and click **Debug** (^D / Shift+F9)

▶ 3. Send a test event

Produce a single JSON message to the `user_events` topic to trigger the breakpoint:

```
make produce-user-events-record
```

▶ 4. Debug

Your IDE will pause at your breakpoint. You can inspect `input`, `state`, and local variables, step through the session logic, and watch `state.sessionId` and `state.eventCount` update as you step over lines.

3.1.1.2 Debugging the `row-driven` PTF using row semantics (`OrderLineExpander`)

> For the full deep-dive, see Remote Debugging `row-driven` Flink PTF UDFs (§1.3).
>
> The `OrderLineExpander` ships in the **same uber JAR** as `UserEventEnricher`, so the same `make deploy-cp-ptf-udf-row-driven` command and the same **"Attach to Flink TaskManager (`row-driven`)"** debug configuration are used. The deploy script registers both PTFs as separate Flink SQL functions and starts an `INSERT INTO orders_expanded SELECT ... FROM TABLE(order_line_expander(input => TABLE orders))` pipeline alongside the user-event enrichment job.

Deploy first: `make deploy-cp-ptf-udf-row-driven`, and then:

▶ 1. Set a breakpoint

Open `OrderLineExpander.java` and click in the gutter at the first line of the `eval()` method:

```
String orderId  = input.getFieldAs("order_id");
```

Or, to inspect the per-item emission, set a breakpoint inside the expansion `for` loop on the `collect(Row.of(...))` call.

▶ 2. Attach the debugger

Use the **same "Attach to Flink TaskManager (Row-Driven)"** configuration as `UserEventEnricher` — both PTFs run in the same TaskManager pod from the same JAR. The IDE will automatically port-forward to the TaskManager pod and attach to the JDWP agent on port `5005`.

- **VS Code:** Open the **Run and Debug** panel (⇧⌘D), select the configuration from the dropdown, and press **F5**
- **IntelliJ IDEA:** Open the **Run/Debug Configurations** dropdown (top-right toolbar), select the configuration, and click **Debug** (^D / Shift+F9)

▶ 3. Send a test order

Produce a single JSON order to the `orders` topic to trigger the breakpoint:

```
make produce-orders-record
```

The sample record has three items in its comma-separated list, so `eval()` will fire once and the `for` loop inside it will emit three rows.

▶ 4. Debug

Your IDE will pause at your breakpoint. Inspect `input`, the parsed `itemParts` and `quantityParts` arrays, and step through the `for` loop watching `i`, `itemName`, `qty`, and the `collect()` call. Notice that:

- There is **no `state` parameter** — row semantics forbids `@StateHint`, so nothing is preserved between rows.
- There is **no `Context` parameter** — no timers or keyed-state services are accessible.
- A single `eval()` call emits **multiple output rows** via repeated `collect()` calls — this is the canonical one-to-many pattern that distinguishes row-semantic PTFs from scalar UDFs.

> **Row-semantic debugging tip:** Because `OrderLineExpander` is stateless, every input row hits `eval()` independently and the framework is free to distribute rows across virtual processors. If you produce multiple orders in quick succession your breakpoint may fire on any TaskManager slot — and rows from different orders may interleave in arbitrary order. Use the `order_id` field in the watch panel to keep track of which row you're inspecting.

3.1.1.3 Debugging the `timer-driven` PTFs (`SessionTimeoutDetector`, `AbandonedCartDetector`, `PerEventFollowUp`, and `SlaMonitor`)

> For the full deep-dive, see Remote Debugging `timer-driven` Flink PTF UDFs.

Deploy first: `make deploy-cp-ptf-udf-timer-driven`, and then:

▶ 1. Set a breakpoint

Open `SessionTimeoutDetector.java` and click in the gutter at the first line of the `eval()` method:

```
String eventType = input.getFieldAs("event_type");
```

Or, to debug the unnamed timer UDF, open `PerEventFollowUp.java` and set a breakpoint at:

```
String eventType = input.getFieldAs("event_type");
```

Or, to debug the Abandoned Cart Detector, open `AbandonedCartDetector.java` and set a breakpoint at:

```
String action = input.getFieldAs("action");
```

Or, to debug the SLA Monitor, open `SlaMonitor.java` and set a breakpoint at:

```
String status = input.getFieldAs("status");
```

Or, to debug a timer callback, set a breakpoint in `onTimer()` of any UDF.

▶ 2. Attach the debugger

Select the **"Attach to Flink TaskManager (Timer-Driven)"** configuration.

- **VS Code:** Open the **Run and Debug** panel (⇧⌘D), select the configuration from the dropdown, and press **F5**
- **IntelliJ IDEA:** Open the **Run/Debug Configurations** dropdown (top-right toolbar), select the configuration, and click **Debug** (^D / Shift+F9)

▶ 3. Send a test event

Produce a single JSON message to the `user_activity` topic to trigger the breakpoint:

```
make produce-user-activity-record
```

▶ 4. Debug

Your IDE will pause at your breakpoint. Inspect `input`, `state`, and local variables, step through the timer registration logic, and watch `state.eventCount` and `state.lastEventType` update as you step over lines.

> **Timer debugging tip:** Timers fire when the watermark advances past the timer's registered time. While paused at a breakpoint, watermarks don't advance, so `onTimer()` won't fire until you resume execution and let the watermark progress. For the unnamed timer UDFs (`PerEventFollowUp` and `SlaMonitor`), note that `onTimer()` fires once per event — not once per partition key. Both the `AbandonedCartDetector` and `SlaMonitor` demonstrate conditional output: `onTimer()` only emits if the cart wasn't checked out or the request wasn't resolved, respectively.

4.0 Resources

4.1 Confluent for Kubernetes (CfK)

- Manage Confluent Platform with Confluent for Kubernetes
- Minikube

4.2 Confluent Platform for Apache Flink

- Stream Processing with Confluent Platform for Apache Flink
- Architecture and Features of Confluent Platform for Apache Flink
- Get Started with Confluent Platform for Apache Flink

4.3 Confluent Cloud for Apache Flink (CCAF)

- Stream Processing with Confluent Cloud for Apache Flink
- Get Started with Confluent Cloud for Apache Flink